Topic 1

Foundations of ESS

1.1 Environmental Value Systems

- History of the Environmental Movement
  - Disasters, literature, the media, int'l agreements
  - Eg. Chernobyl (1986) & Fukushima (2011)
  - Eg. Gaia (1979, James Lovelock)
    - The earth is a living organism, w/ humans upsetting balance
  - Eg. *An Inconvenient Truth* (2006) & stewardship
  - Eg. Stockholm Declaration (1972), Rio Summit (1992), Kyoto Protocol (1997)
    - Our Common Future Report (1987) seeking sustainable development through int'l collaboration & adoption of Agenda 21
    - Indirect ecosystem values, changed perceptions on economic growth

Example 1 - Gulf of Mexico Oil Spill

- Deepwater Horizon oil rig operated by offshore company Transocean
- Largest marine oil spill in history
- Explosion on Apr 20, capped on July 15
- Revisions to offshore drilling regulations
- Public outcry

Example 2 - Rachel Carson's *Silent Spring*

- Published 1962 by conservationist, marine biologist and author Rachel Carson
- Documentation of harmful effects of pesticides along food chains
- Widespread concern about pollutants in the environment
- Reversal in national pesticide policy
- Ban on DDT
- Inspired the environmental movement

Example 3 - Earth Day

- April 22 (began 1970) by Gaylord Nelson
- Channelling anti-war protests
- Global encouragement of environmental campaigns
- Movement of environmental issues onto the national political agenda
- Now celebrated by more than 1 billion people a year

1. 1 Environmental Value Systems

- A worldview that shapes the way a group perceives and evaluates environmental issues
- Can be seen as a system
  - Inputs: education, SEEP factors, the media
  - Outputs: perspectives, decisions, action
  - Flows: process or transformation of information

EVSs

Ecocentric

> the belief that ecology is central to humanity

> emphasises a less materialistic way of life

> sees nature as having intrinsic value for humanity

> focuses on biorights

Anthropocentric

> the belief that humans should sustainably manage the global system

> through taxes, regulation & legislation, debate

> exploitation can continue given adjustments

> importance of education & politics

Technocentric

> the belief that technological advancements can provide solutions to environmental problems

> optimistic, research encouraged to understand systems

> faith in humanity's ability to overcome difficulties

1. 2 Systems & Models

- System: a collection of parts and the relationships between them, coming together as a whole
- Reductionist vs. systems approach
  - Dividing & studying separately
  - Holistic, visualising complex interactions
- All systems have: inputs, outputs, storages, flows (+ boundaries)
  - Boxes and arrows
- Transfers or transformations

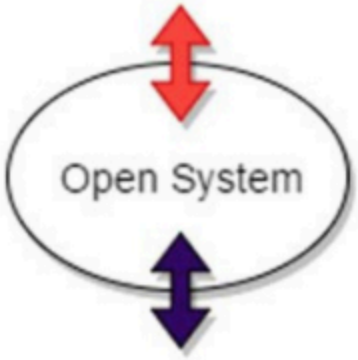
Types of Systems

Open

> both energy & matter exchanged

> feedback & adjustment

> organic

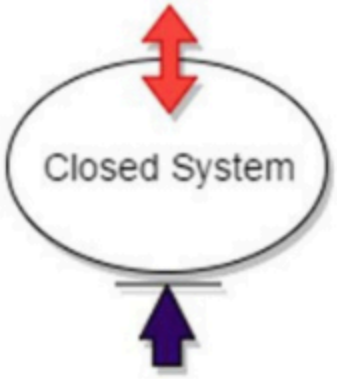


Closed

> energy is exchanged; matter is not

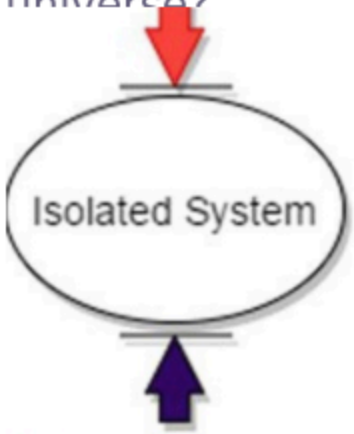
> uncommon

> Earth



Isolated

> neither matter nor energy is exchanged

> not naturally occuring

> universe?